

This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the —README.md— for this assignment includes instructions to regenerate this handout with your typeset L^AT_EX solutions.

2.a Here we will calculate the gradient of the loss of the reward estimate of a state action pair $\hat{r}(o, a)$ as approximated by a function $\phi_w(o, a)$ defined by parameters w . The gradient of the loss will be taken with respect to w .

First off, the softmax function with two arguments can be expressed as a sigmoid. For each of the probability functions within the Bradley-Terry preference model we see that

$$\hat{P}[\sigma^1 \succ \sigma^2] = \frac{\exp(\sum_t \hat{r}(o_t^1, a_t^1))}{\exp(\sum_t \hat{r}(o_t^1, a_t^1)) + \exp(\sum_t \hat{r}(o_t^2, a_t^2))} \quad (1)$$

$$= \left[1 + \frac{\exp(\sum_t \hat{r}(o_t^2, a_t^2))}{\exp(\sum_t \hat{r}(o_t^1, a_t^1))} \right]^{-1} \quad (2)$$

$$= \left[1 + \exp\left(-\sum_t (\hat{r}(o_t^1, a_t^1) - \hat{r}(o_t^2, a_t^2))\right) \right]^{-1}. \quad (3)$$

Similarly,

$$\hat{P}[\sigma^2 \succ \sigma^1] = \left[1 + \exp\left(-\sum_t (\hat{r}(o_t^2, a_t^2) - \hat{r}(o_t^1, a_t^1))\right) \right]^{-1}. \quad (4)$$

If we substitute $\phi_w(o, a)$ for $\hat{r}(o, a)$ within $\hat{P}[\sigma^1 \succ \sigma^2]$ and $\hat{P}[\sigma^2 \succ \sigma^1]$ of the Bradley-Terry preference model, we find

$$\hat{P}[\sigma^1 \succ \sigma^2] = \left[1 + \exp\left(-\sum_t (\phi_w(o_t^1, a_t^1) - \phi_w(o_t^2, a_t^2))\right) \right]^{-1} \quad (5)$$

$$\hat{P}[\sigma^2 \succ \sigma^1] = \left[1 + \exp\left(-\sum_t (\phi_w(o_t^2, a_t^2) - \phi_w(o_t^1, a_t^1))\right) \right]^{-1}. \quad (6)$$

To encapsulate and simplify the notation for working through the gradient in subsequent steps, let's define

$$s_1 = \sum_t \phi_w(o_t^1, a_t^1) \quad (7)$$

$$s_2 = \sum_t \phi_w(o_t^2, a_t^2) \quad (8)$$

so equations 5 and 6 may now be respectively expressed as

$$\hat{P}[\sigma^1 \succ \sigma^2] = [1 + \exp(-(s_1 - s_2))]^{-1} \quad (9)$$

$$\hat{P}[\sigma^2 \succ \sigma^1] = [1 + \exp(-(s_2 - s_1))]^{-1}. \quad (10)$$

Taking the gradient with respect to the parameters w of the estimating function ϕ_w we find

$$\nabla_w \hat{P}[\sigma^1 \succ \sigma^2] = \frac{\exp(-(s_1 - s_2))}{[1 + \exp(-(s_1 - s_2))]^2} \nabla_w (s_1 - s_2) \quad (11)$$

$$\nabla_w \hat{P}[\sigma^2 \succ \sigma^1] = \frac{\exp(-(s_2 - s_1))}{[1 + \exp(-(s_2 - s_1))]^2} \nabla_w (s_2 - s_1). \quad (12)$$

We will substitute these two gradient expressions into the expression for the gradient of the loss function, $\nabla_w \text{loss}$, in steps to follow. Now, at this stage we can express top-level differentiation of the loss as

$$\nabla_w \text{loss} = - \sum_{(\sigma^1, \sigma^2, \mu) \in \mathcal{D}} \mu(1) \nabla_w \log \hat{P}[\sigma^1 \succ \sigma^2] + \mu(2) \nabla_w \log \hat{P}[\sigma^2 \succ \sigma^1] \quad (13)$$

and the gradients of the terms inside the sum are of the form

$$\nabla_w \log \hat{P}[\sigma^1 \succ \sigma^2] = \frac{\nabla_w \hat{P}[\sigma^1 \succ \sigma^2]}{\hat{P}[\sigma^1 \succ \sigma^2]} \quad (14)$$

$$\nabla_w \log \hat{P}[\sigma^2 \succ \sigma^1] = \frac{\nabla_w \hat{P}[\sigma^2 \succ \sigma^1]}{\hat{P}[\sigma^2 \succ \sigma^1]}. \quad (15)$$

So if we substitute 9 and 11 into 14 we find

$$\nabla_w \log \hat{P}[\sigma^1 \succ \sigma^2] = \frac{\exp(-(s_1 - s_2)) [1 + \exp(-(s_1 - s_2))]}{[1 + \exp(-(s_1 - s_2))]^2} \nabla_w (s_1 - s_2) \quad (16)$$

$$= \frac{\exp(-(s_1 - s_2)) \nabla_w (s_1 - s_2)}{[1 + \exp(-(s_1 - s_2))]} \frac{\exp((s_1 - s_2))}{\exp((s_1 - s_2))} \quad (17)$$

$$= \frac{-\nabla_w (s_2 - s_1)}{1 + \exp(-[s_2 - s_1])}. \quad (18)$$

By a similar series of operations

$$\nabla_w \log \hat{P}[\sigma^2 \succ \sigma^1] = \frac{-\nabla_w (s_1 - s_2)}{1 + \exp(-[s_1 - s_2])}. \quad (19)$$

By substituting equalities 18 and (19) into (13) we can now express the gradient of the loss as

$$\nabla_w \text{loss} = \sum_{(\sigma^1, \sigma^2, \mu) \in \mathcal{D}} \mu(1) \frac{\nabla_w (s_2 - s_1)}{1 + \exp(-[s_2 - s_1])} + \mu(2) \frac{\nabla_w (s_1 - s_2)}{1 + \exp(-[s_1 - s_2])}. \quad (20)$$

Finally, re-inserting equalities for s_1 and s_2 into 20 we are left with the final expression

$$\nabla_w \text{loss}(\phi_w) = \sum_{(\sigma^1, \sigma^2, \mu) \in \mathcal{D}} \mu(1) \frac{\nabla_w (\sum_t \phi_w(o_t^2, a_t^2) - \sum_t \phi_w(o_t^1, a_t^1))}{1 + \exp(-[\sum_t \phi_w(o_t^2, a_t^2) - \sum_t \phi_w(o_t^1, a_t^1)])} + \quad (21)$$

$$\mu(2) \frac{\nabla_w (\sum_t \phi_w(o_t^1, a_t^1) - \sum_t \phi_w(o_t^2, a_t^2))}{1 + \exp(-[\sum_t \phi_w(o_t^1, a_t^1) - \sum_t \phi_w(o_t^2, a_t^2)])} \quad (22)$$

which may be simplified as

$$\nabla_w \text{loss}(\phi_w) = \sum_{(\sigma^1, \sigma^2, \mu) \in \mathcal{D}} \mu(1) \frac{\sum_t [\nabla_w \phi_w(o_t^2, a_t^2) - \nabla_w \phi_w(o_t^1, a_t^1)]}{1 + \exp(-\sum_t [\phi_w(o_t^2, a_t^2) - \phi_w(o_t^1, a_t^1)])} + \quad (23)$$

$$\mu(2) \frac{\sum_t [\nabla_w \phi_w(o_t^1, a_t^1) - \nabla_w \phi_w(o_t^2, a_t^2)]}{1 + \exp(-\sum_t [\phi_w(o_t^1, a_t^1) - \phi_w(o_t^2, a_t^2)])} \quad (24)$$

by the linearity of the sum operator and since

$$\nabla_w \left(\sum_t \phi_w(o_t, a_t) \right) = \sum_t \nabla_w \phi_w(o_t, a_t). \quad (25)$$

Noticing that in each of the terms, one over the denominator constitutes a sigmoid function, the gradient of the loss in terms of w may be more neatly expressed as

$$\nabla_w \text{loss}(\phi_w) = \sum_{(\sigma^1, \sigma^2, \mu) \in \mathcal{D}} \mu(1) \cdot \text{Sigmoid} \left(\sum_t [\phi_w(o_t^2, a_t^2) - \phi_w(o_t^1, a_t^1)] \right) \sum_t [\nabla_w \phi_w(o_t^2, a_t^2) - \nabla_w \phi_w(o_t^1, a_t^1)] + \quad (26)$$

$$\mu(2) \cdot \text{Sigmoid} \left(\sum_t [\phi_w(o_t^1, a_t^1) - \phi_w(o_t^2, a_t^2)] \right) \sum_t [\nabla_w \phi_w(o_t^1, a_t^1) - \nabla_w \phi_w(o_t^2, a_t^2)]. \quad (27)$$

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